1. lit2026 W11: generic-method engineering — results

Workstream W11 of the lit2026 speedup program, 2026-10-11; stacked on aburan28/crypto#1681. Run `run-20261011T2049Z`; host Apple M4 Pro (14 cores, shared, load 72–155), **isolation L0**. Arms: `original` = `4b6e09b78`; `previous/candidate` = this branch with its switch off/on. Deterministic counters are the gate; wall ratios are ratio of medians with [min, max] of per-round ratios. No exponent, deployed-curve or IC-versus-rho claim.

item (paper)	built	verdict
(a) ePrint 2026/2102	`bsgs_fast` kernels `paired`, `coz` (`BSGS_STEP`)	speedup: paired 1.15–1.67× wall vs original, 40–56 bits; 16.75 ⟶ 3.71 M-eq per point
(b) ePrint 2026/704	x-only binary evaluators; one-inversion pair step (`BTW_XEVAL`)	step 1.50× fewer `F_(q^8)` mults, 1.29× wall; class walk end to end: no change
(c) ePrint 2025/1652	cubical ladder for the k = 2 Tate pairing	16–19% fewer `F_p` M-eq than Jacobian Miller (28–52-bit r); identical values
(d) ePrint 2026/1823	DP trail recording in `koblitz_strong_rho`	diagnostic: competing-risk geometric law; fruitless abandonment truncates trails at dp = 12
(e) arXiv 2508.14011	Algorithm 3 in Rust + `prime_fast` rho	diagnostic: 1.077 ± 0.020 × √(πn/4) over 720 solves; paper's 8/12/16-bit cards use an ascending search

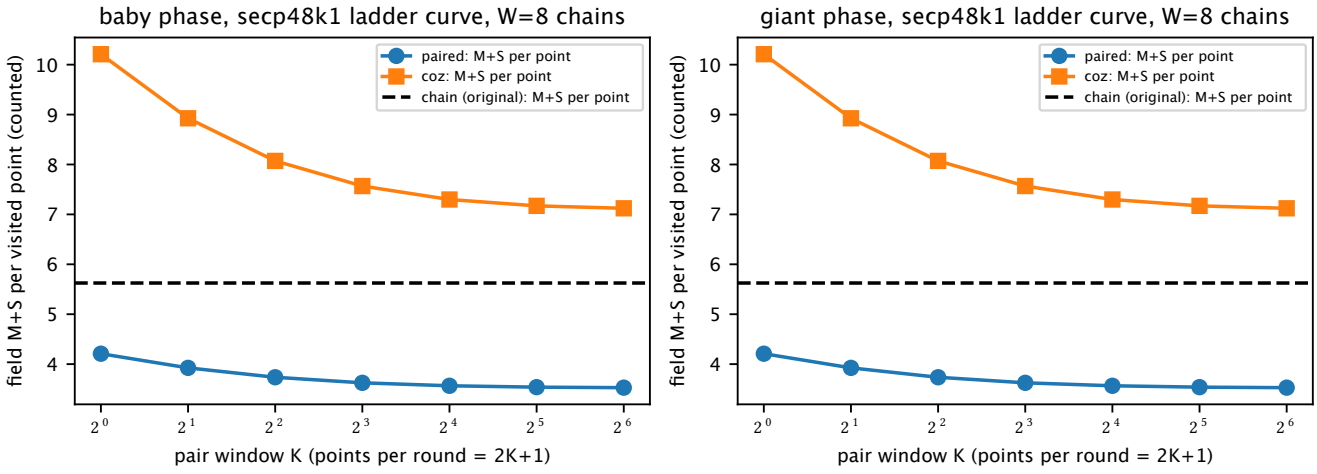
Papers: abstracts of 2026/2102, 2026/704, 2025/1652, 2026/1823 (PDFs returned HTTP 403 from this host; formulas here are our own derivations), full PDF of 2508.14011.

2. (a) BSGS: co-Z / paired giant steps, windowed batch inversion

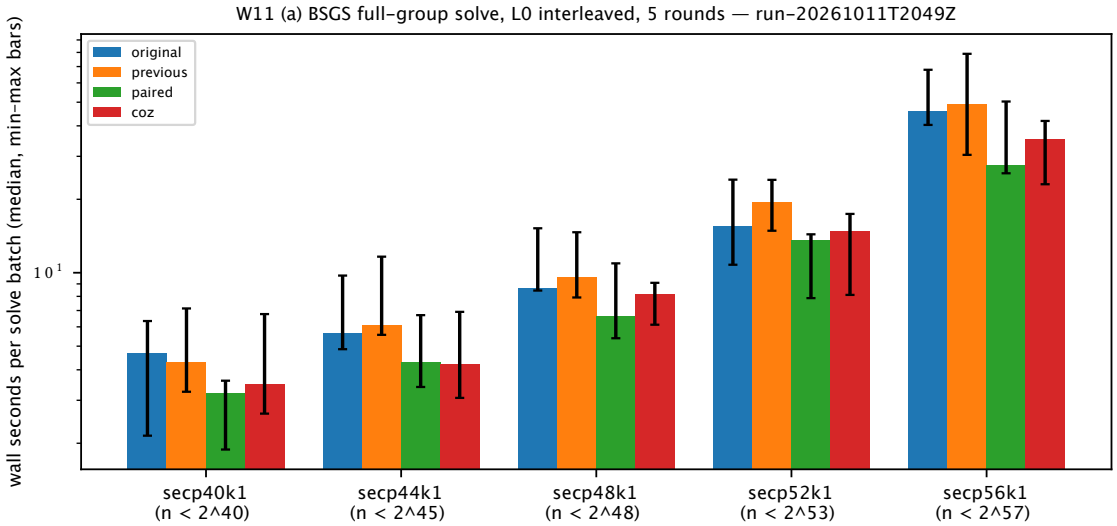
`paired`: affine centre C, offsets jA (j ≤ K), C ± jA share 1/(x(jA) − x(C)); non-centre points x-only; one batch inversion over W(K+1) denominators. `coz`: Jacobian centre, co-Z conjugate pairs, Z's inverted per window (our reading of 2026/2102).

curve	kernel	K	M/pt	S/pt	I/pt	M-eq/pt
secp48k1	chain	–	4.625	1.000	0.1250	16.75
secp48k1	paired	32	2.539	1.000	0.00193	3.71
secp48k1	coz	32	5.117	2.054	0.00193	7.34
secp56k1	chain	–	4.625	1.000	0.1250	18.88
secp56k1	paired	32	2.539	1.000	0.00193	3.74

W11 (a) BSGS step kernels: counted field multiplications+squarings per point — run-20261011T2049Z



curve	orig. s	prev. s	paired s	coz s	orig/paired	prev/paired	orig/prev
secp40k1	4.67	4.28	3.20	3.49	1.459 [1.054, 1.917]	1.337 [1.223, 2.037]	1.091 [0.655, 1.091]
secp44k1	5.63	6.12	4.27	4.22	1.316 [1.015, 2.002]	1.430 [1.262, 1.737]	0.920 [0.804, 1.399]
secp48k1	8.63	9.55	6.64	8.16	1.301 [1.299, 1.603]	1.439 [1.203, 1.849]	0.904 [0.847, 1.157]
secp52k1	15.55	19.40	13.58	14.77	1.145 [0.996, 1.976]	1.429 [1.204, 2.466]	0.801 [0.726, 1.002]
secp56k1	45.84	48.99	27.44	35.32	1.670 [1.349, 1.670]	1.785 [1.190, 1.849]	0.936 [0.860, 1.327]

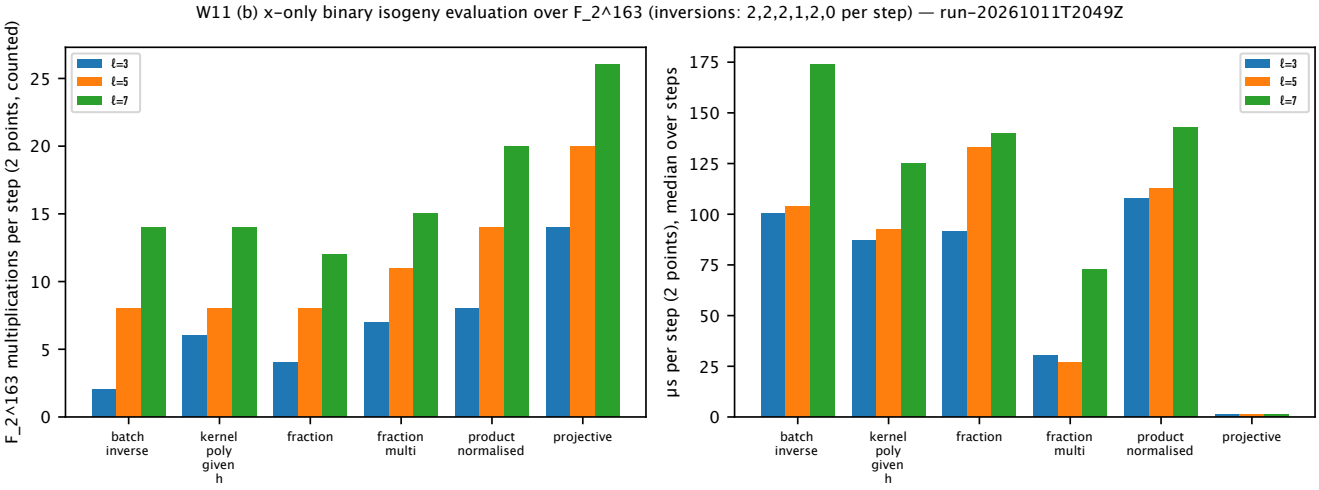


Answers and table sizes identical across arms. The `previous` arm (same binary, chain kernel, now generic over a field trait) ran 0.80–0.94× of `original` at four sizes, with intervals crossing 1; recorded, not explained.

3. (b) x-only binary isogeny evaluation

Counted `F_(2^163)` mul / inv per step (two points); all evaluators equal `velu_x_map`, the Vélu point map's abscissa lies on the `velu_codomain` curve.

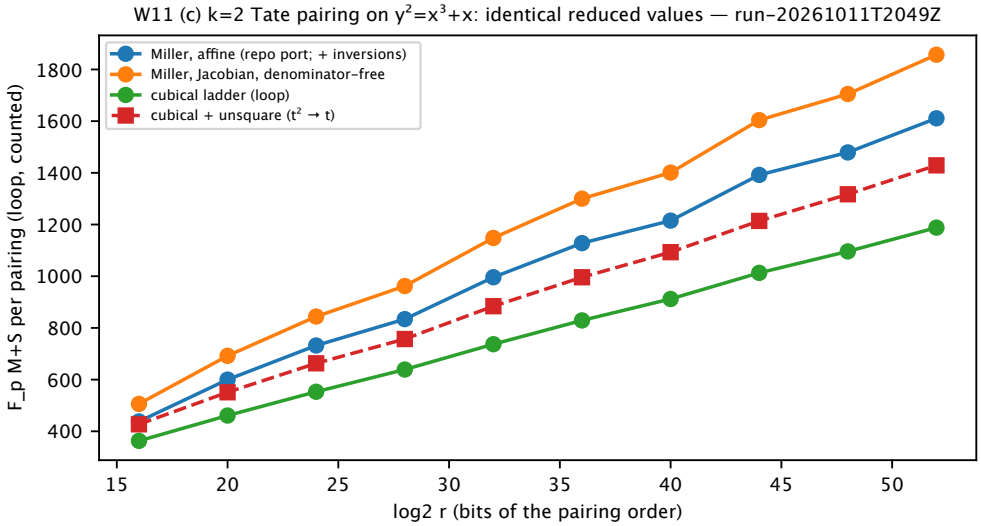
ℓ (s)	batch inv.	kernel poly	fraction	fraction, 1 inv.	product	projective
3 (1)	2 / 2	6 / 2	4 / 2	7 / 1	8 / 2	14 / 0
5 (2)	8 / 2	8 / 2	8 / 2	11 / 1	14 / 2	20 / 0
7 (3)	14 / 2	14 / 2	12 / 2	15 / 1	20 / 2	26 / 0



The product form $x \cdot \Pi((x\,x_i + \sqrt{b})/(x + x_i))^2$ equals $(\Pi\, x_i)^2 \cdot \varphi(x)$ (checked on every kernel). Torsion-walk step at (19, 1), $\ell = 457$ in `F_(q^8)`: 627 460 ⟶ 419 070 mults, 916 ⟶ 458 inversions, 1564 ⟶ 1211 µs per kernel. End-to-end class walk (12 rounds, identical snapshots): walk seconds 1.280 / 1.164 / 1.198 (original / previous / candidate), original/candidate 1.069 [0.714, 1.831]: **no change**.

4. (c) k = 2 pairing: cubical ladder vs Miller

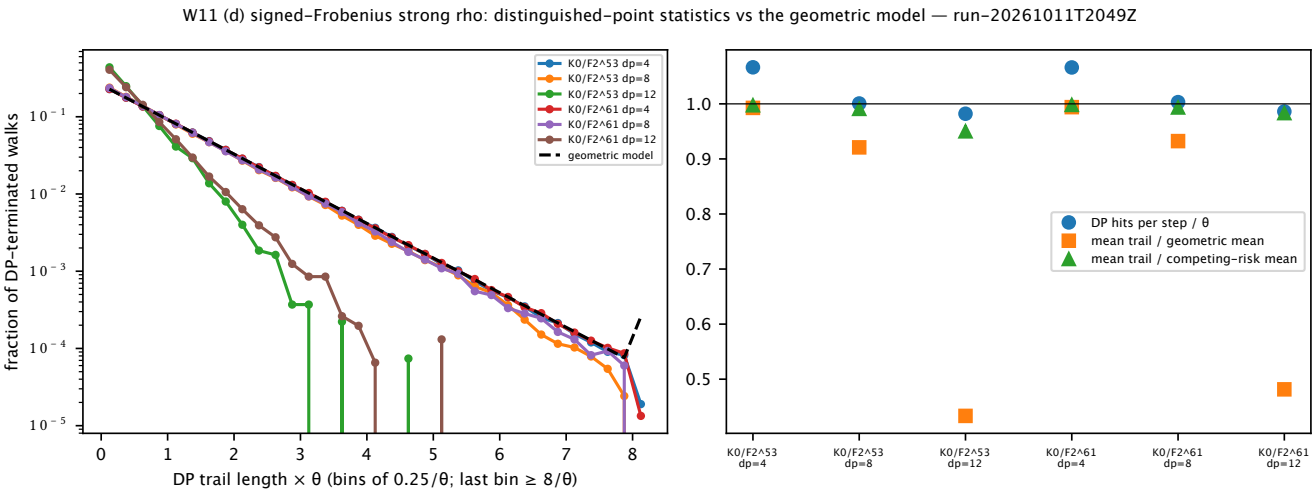
r bits	affine Miller M+S / I	Jacobian Miller M+S	cubical M+S / I	unsquare M+S	(cub.+unsq.)/Jac. M+S	same, M-eq
16	438 / 52	506	363 / 1	64	0.844	0.909
32	996 / 120	1148	737 / 1	147	0.770	0.830
40	1215 / 146	1401	912 / 1	181	0.780	0.840
52	1611 / 194	1857	1188 / 1	241	0.770	0.828



The cubical ladder's reduced value is the square of the pairing on every instance; `unsquare` (exponent (r+1)/2, counted) restores it. Values identical to Miller and to `eds_mov::modified_tate`.

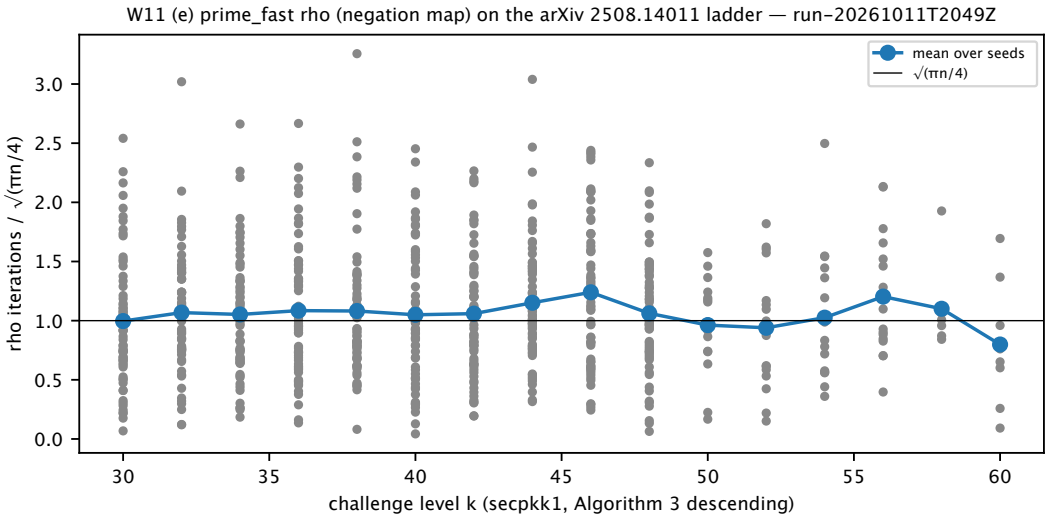
5. (d) Distinguished-point statistics

curve	dp	DPs	hits/θ	trail/geo	trail/c-r	hazard	KS geo / c-r / 95%	steps/ideal
K0/F2^53	4	1 945 590	1.066	0.993	0.997	2.97e-4	0.0023 / 0.0006 / 0.0010	0.82
K0/F2^53	8	165 190	1.001	0.921	0.991	2.96e-4	0.031 / 0.0042 / 0.0033	1.18
K0/F2^53	12	13 518	0.982	0.433	0.950	2.91e-4	0.301 / 0.022 / 0.012	1.58
K0/F2^61	4	2 465 006	1.066	0.994	0.998	2.55e-4	0.0019 / 0.0005 / 0.0009	0.80
K0/F2^61	8	183 269	1.003	0.932	0.993	2.56e-4	0.026 / 0.0025 / 0.0032	1.01
K0/F2^61	12	15 276	0.986	0.482	0.983	2.54e-4	0.264 / 0.0088 / 0.011	1.37



6. (e) The secpkk1 challenge ladder

Cards 6/24/32/48 = Algorithm 3 as printed (descending); cards 8/12/16 = ascending search from 2^(k−1). 720 verified solves at 30–60 bits: iterations/√(πn/4) mean 1.077 (s.d. 0.546, s.e. 0.020), median 0.998; DP start-up overhead estimate 0.12–0.18.



7. Limits

L0 on a heavily shared host; (a) is not an ecbench method-level comparison (ecbench's BSGS arms do not call `bsgs_fast`), and the rho walk does not share the BSGS step; (b) scanned sect163-field curves, $s \le 3$, paper formulas unavailable; (c) u64 fields only, squared pairing corrected by `unsquare`; (d) two curves, one walk configuration; (e) 8–16 seeds per level above 48 bits.